

Metabolic syndrome and kidney disease

Holly J. Kramer¹ and Shivam Joshi²

¹Departments of Medicine and Public Health Sciences, Loyola University Chicago, Maywood, Maywood, IL, United States ²Division of Nephrology, Department of Medicine, New York University, Langone Health, New York, NY, United States

Introduction

Humans have evolved over the last centuries when food was not always available, and such nutritional challenges created efficiency in storing excess calories and limiting energy expenditure. In 1962 Dr. Neel described a metabolic process whereby humans adapt to dynamics in caloric intake and named this evolutionary adaptation the thrifty genome [1]. As the body stores the excess intake of calories that exceed energy expenditure, insulin resistance ensues. While a thrifty genome was helpful with food scarcity, efficiency of caloric storage can be now viewed as maladaptive if food scarcity no longer exists and caloric intake is instead excessive. In this chapter, we review metabolic syndrome and its origins and the organ damage that often accompanies metabolic syndrome, including chronic kidney disease (CKD). In this chapter, you will learn the cellular consequences of insulin resistance and postulated mechanisms that mediate kidney disease in the setting of metabolic syndrome.

Metabolic syndrome and its origins

Metabolic syndrome is a nutritional disorder rooted in the overconsumption of calories that exceeds energy expenditure leading to overweight and obesity. Obesity, defined as a body mass index (BMI) greater than or equal to 30 kg/m², continues to be a growing epidemic within the United States and globally. Over 93 million United States adults are now obese but the prevalence differs by age and racial/ethnicity. Within age-groups the highest prevalence is among men (40.8%) and women (44.7%) age 40–59 years old and

lowest among men (34.8%) and women (36.5%) age 20–39 years [2]. Across racial/ethnic groups, obesity prevalence ranges from as low as 10.1% and 14.8% among Asian men and women, respectively, to as high as 43.1% and 50.6% among Hispanic men and women, respectively. Across the globe, obesity affects approximately 650 million people while over 1.3 billion adults are overweight, defined as a BMI between 25 and 29.9 kg/m² [3]. These statistics are based on BMI and can be misleading. Metabolic syndrome can occur with increased visceral adiposity known as android obesity that itself increases cardiovascular disease (CVD) and diabetes risk. Many argue that waist circumference or waist-to-hip ratio should be used to define obesity instead of BMI because of the importance of visceral adiposity. In addition, due to racial/ethnic differences in body size, racial/ethnic-specific definitions for both BMI and waist circumference have also been suggested.

The epidemic of obesity around the globe has been fueled by a ubiquity of calorically dense, highly palatable processed foods but not everyone has been affected. Differences in obesity risk have been attributed to numerous genetic variants across the genome, but these discovered variants each contribute very small effects on energy intake or energy expenditure. Thus the overwhelming majority of obesity is considered polygenetic [4] and a complex disease meaning that obesity occurs due to the interaction of multiple genetic factors that determines susceptibility to obesity combined with an obesogenic environment (access to densely caloric foods and sedentary lifestyle). Rarely, obesity is attributed to a single genetic variant (monogenic obesity) or to several defined genetic variants/defects or chromosomal abnormalities; these obesity

cases are usually accompanied by developmental abnormalities and termed syndromic obesity.

Regardless of genetic background, weight gain will not occur without energy intake exceeding energy expenditure. Weight gain is essential for development of metabolic syndrome but it is not sufficient because where the excess calories are stored is just as important as the weight gain itself. In contrast to overall obesity risk that is mainly mediated by nongenetic factors, the distribution patterns of adiposity are strongly familial, and genetically determined as demonstrated by strong sex differences in adiposity distribution. In some persons the excess adipose tissue may be readily stored in hips, thighs, and buttocks (peripheral), while in others the adipose tissue is mainly stored in the abdomen, liver, and muscles (truncal). Over 80% of body fat is normally stored in the subcutaneous adipose tissues while the other 20% is stored in visceral adipose tissue (omentum, mesentery, etc.) [5]. In some individuals, excess caloric intake is stored as fat preferentially in visceral adipose tissue and this is largely determined by genetic factors. These genetic factors that influence body fat distribution also account for sex differences in body fat distribution [6]. This adipose tissue stored in the abdomen or visceral adipose tissue leading to android obesity primarily drives the development of metabolic syndrome, or the clustering of major cardiovascular risk factors, including hypertension, dyslipidemia, and glucose intolerance.

In addition to where the fat is distributed, the overall ability to store excess fat also varies across individuals. The ability to store excess calories as defined by the total number of adipose cells is also largely genetic but can be influenced during childhood via being overweight. The number of adipose cells increases during childhood but then, after adolescence, remains constant during adulthood [7]. Thus regardless of weight loss or weight gain, the total number of adipose cells generally stays the same during the adult life span of an individual. Adipose cells do demonstrate cell turnover at a rate of 10% annually, but this turnover does not differ by obesity status and does not alter the total number of adipose cells available for storage [7]. When storage capacity of existing adipocytes is exceeded, fat must then be stored in the liver and muscle. Thus metabolic syndrome can be viewed as the immersion of a genetically susceptible individual combined with excess caloric intake.

Metabolic syndrome and its definition

In 1988 the presence of several metabolic traits was noted to associate with a heightened risk of type 2 diabetes and CVD. This clustering of elevated glucose levels, hyperinsulinemia, high levels of very-low-density

lipoprotein triglycerides, and low-high-density lipoprotein cholesterol was entitled Syndrome X by Dr. Reaven [8] and then later changed to Metabolic Syndrome [9,10]. Increased waist circumference was later added as an additional metabolic syndrome trait by Norman Kaplan [11]. The Third Report of the Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III, or ATP III) from the National Cholesterol Education Program (NCEP) later established a definition for metabolic syndrome. The presence of at least three of the following five criteria defined metabolic syndrome: central obesity (waist circumference in men >102 cm and in women >88 cm), hypertriglyceridemia (≥ 150 mg/dL), low-high-density lipoprotein (HDL) cholesterol (men <40 mg/dL, women, <50 mg/dL), elevated fasting glucose (≥ 110 mg/dL), and elevated blood pressure (BP) ($\geq 130/85$ mmHg) (see Table 39.1) [12]. The World Health Organization (WHO) also developed a definition of metabolic syndrome that tied factors associated with insulin resistance such as obesity, dyslipidemia, and hypertension. The WHO definition also allowed persons with diabetes to be diagnosed with metabolic syndrome but the diagnostic criteria could not be readily applied in clinical settings. The European Group for Study of Insulin Resistance postulated that insulin resistance was central to metabolic syndrome and used elevated fasting insulin (<75 th percentile) to define insulin resistance but this meant persons with diabetes could not be diagnosed. The European group also eliminated increased urine albumin excretion as a metabolic syndrome trait given its lack of specificity with insulin resistance [13].

The use of hyperinsulinemia to define metabolic syndrome is troublesome because of heterogeneity in measurement, differing classification schemes, and lack of standardization across studies [9]. Many studies define hyperinsulinemia as insulin levels greater than the 75th percentile for the general population but this particular definition can be misleading because this assumes a threshold dependence for insulin resistance. Similar to BP, it is likely that insulin resistance is a continuous measure where even low levels of insulin resistance can lead to endothelial dysfunction and organ dysfunction in susceptible individuals. In addition, levels of insulin that indicate resistance likely vary substantially across individuals just as the association of obesity measures with insulin resistance varies across individuals. Subsequently, the NCEP and the scientific statement from the American Heart Association and the National Heart, Lung, and Blood Institute [9] eliminated insulin resistance or obesity as a required trait (Table 39.1). Instead, at least three of five potential criteria need to be present to

TABLE 39.1 Metabolic syndrome definitions.

	Insulin resistance	Elevated glucose	Low HDL	Elevated triglycerides	Overweight	Elevated blood pressure	Urine albumin excretion
¹² NCEP-3 + traits	Not required, not a trait	≥ 100 mg/dL	<40 mg/dL (men), <50 mg/dL (women) or medication use	≥ 150 mg/dL or medication use	WC >102 cm (men), >88 cm (women)	≥ 130/85 mmHg or medication use	Not a trait
¹²² WHO-2 + traits	Required: top 25th percentile in insulin levels or fasting glucose ≥ 110 mg/dL or 2 h glucose ≥ 140 mg/dL	See insulin resistance	<35 mg/dL (men); <40 mg/dL (women)	≥ 150 mg/dL	WHR >0.9 (men), >0.85 (women), or BMI ≥ 30 kg/m ²	≥ 140/90 mmHg	> 20 µg/min
¹²³ EGIR-2 + traits	Required: op 25th percentile in insulin levels	≥ 110 mg/dL	<40 mg/dL	≥ 180 mg/dL	WC ≥ 94 cm (men), ≥ 80 cm (women)	≥ 140/90 mmHg	Not a trait
¹²⁴ IDF-2 or more traits	Not required	≥ 100 mg/dL	<40 mg/dL (men); <50 mg/dL	≥ 150 mg/dL or on medications	Required: + WC ≥ 94 cm (men), ≥ 80 cm (women)	≥ 135/85 mmHg or medication use	Not a trait
¹²⁵ AACE	Not required	≥ 110 mg/dL; 2 h glucose ≥ 140 mg/dL	<40 mg/dL (men), <50 mg/dL (women) or medication use	≥ 150 mg/dL	Required: BMI >25 kg/m ² or WC ≥ 102 cm (men), ≥ 88 cm (women)	≥ 130/85 mmHg or medication use	Not a trait

AACE, American Association of Clinical Endocrinologists; ATP-NCEP, Adult Treatment Panel III from the National Cholesterol Education Program; BMI, body mass index; EGIR, European group for the Study of Insulin Resistance; HD + If Asian, different waist circumference thresholds should be utilized: ≥ 90 cm (men) or ≥ 80 cm (women); IDF, International Diabetes Federation; WC, waist circumference; WHO, World Health Organization; WHR, waist-to-hip ratio.

diagnose presence of metabolic syndrome. In contrast, the International Diabetes Foundation required the presence of obesity but not insulin resistance. The main problem with this strategy is that many adults with obesity do not have metabolic syndrome, especially those who are physically active [14]. Currently applied definitions do not account for family history of diabetes or CVD and do not address the organ manifestations of metabolic syndrome such as fatty liver disease and CKD factors associated with obesity and heightened mortality. It should be noted that the constellation of cardiovascular risk factors in adults with obesity was first recorded by a Swedish physician Eskil Kylin in 1923 and was conceptualized to describe [15] the end-organ damage from insulin resistance [15]. This early definition of metabolic syndrome included hyperuricemia, a trait that often predates the development of type 2 diabetes [16] and frequently accompanies CKD.

In 2009 a joint statement from the International Diabetes Federation Task Force on Epidemiology and Prevention and the National Heart, Lung, and Blood Institute along with several other organizations convened a conference to discuss the controversies regarding the terminology and criteria for metabolic syndrome diagnosis [17]. The members of all organizations agreed that metabolic syndrome represents the presence of multiple metabolic risk factors for both CVD and diabetes. However, the impact of obesity on overall health is not limited to the five metabolic syndrome traits defined by the NCEP [12]. Low HDL,

glucose intolerance, increased triglycerides, abdominal adiposity, and elevated BP are traits that heighten CVD risk [9]. Metabolic syndrome also portends heightened risk for diabetes and in most individuals reflects insulin resistance. However, the constellation of metabolic syndrome traits is not always a reflection of insulin resistance and could instead reflect dietary patterns, sedentary lifestyle, aging, imbalance in estrogen and testosterone levels and genetic factors [9]. The ATP III metabolic syndrome definition is highly specific for insulin resistance but its sensitivity for hyperinsulinemia is overall low [18–20] because criteria do not address racial, sex, or age differences in the effects of adiposity on health and the strong genetic determinants of adiposity distribution and susceptibility to metabolic syndrome [21]. In addition, metabolic syndrome itself does not impart a higher predictive value for CVD risk compared to the sum of its individual components and thus the overall value of the metabolic syndrome definition remains questionable [22–24]. Regardless of the controversy, the metabolic syndrome may be viewed as a signal of high risk for diabetes, CVD and CKD, and signals a need for implementing nutritional and exercise strategies to reduce risk.

Metabolic syndrome and comorbidities

Based on the ATP III criteria, the overall prevalence of metabolic syndrome is now approximately 34% for both men and women [25]. The burden of metabolic syndrome is particularly high among Hispanic/Latino

adults. In the Hispanic Community Health Study/Study of Latinos, a cohort study of 16,415 Hispanics from 6 Hispanic/Latino backgrounds, metabolic syndrome was present in over one-third of men and women and increased with age. The highest prevalence of metabolic syndrome was noted among the participants with a Puerto Rican Hispanic/Latino background at 41% [26].

Lifestyle modifications affect all metabolic syndrome risk factors; this supports the hypothesis that the risk factors are not independent of each other and are all mediated by one shared factor likely insulin resistance [9,27]. Insulin resistance combined with adiposity leads to abnormal lipid accumulation in skeletal muscle and the liver. In persons without enough adipose cells, excess calories must be stored elsewhere and this may include the liver. Despite the growth of fatty liver disease termed nonalcoholic fatty liver disease (NAFLD), metabolic syndrome criteria do not include fatty liver infiltration in the absence of heavy alcohol ingestion or NAFLD [12]. NAFLD describes a wide spectrum of disease ranging from fatty infiltration (nonalcoholic steatosis) to nonalcoholic steatohepatitis to cirrhosis and liver failure. While the amount of visceral adipose tissue correlates with the amount of hepatic adiposity, abdominal adiposity itself does not itself predict risk of NAFLD [28]. It appears that both adiposity and insulin resistance are necessary for the development of NAFLD because genetic mutations of insulin receptors that alter insulin resistance can prevent hepatic adiposity [29,30]. While adiposity is not necessary for NAFLD, 90% of NAFLD is associated with insulin resistance [28]. In healthy states the liver contains only 5% lipids in total weight but as the deposition of fat increases, the fatty hepatocyte starts to resemble brown adipose tissue [28]. Thus if excess caloric storage exceeds the capacity of the set number of adipocytes available, the liver acts as an additional storage depot for lipids similar to adipose tissue. In its early stages, NAFLD may be treated thru weight loss [31–33]. However, weight loss must be at least 5% of baseline weight to achieve improvements in liver histology. In one study, participants who lost at least 7% of baseline weight showed improved liver histology and significant improvements in steatosis despite the advanced state of liver disease [32].

Another storage depot for excess calorie storage is muscle. Intramyocellular lipid content is high among highly trained athletes because these fatty acids provide a quick energy source for physical activity [30]. High muscle lipid content in highly active individuals does not reflect insulin resistance. Among sedentary individuals, however, the intramyocellular lipid content correlates strongly with insulin resistance and exceeds the correlation of visceral fat content with insulin resistance [30,34]. Due to the confounding

effects of exercise and the difficulties in accurate measurements, intramyocellular lipid content has not been utilized to quantify disease risk outside of research settings [30].

Chronic kidney disease

Metabolic syndrome is associated with increased risk of CKD defined by increased urine albumin excretion [35,36]. In healthy states, albumin normally accounts for less than 10% of the total urinary protein excretion and usually ranges from 2 to 10 mg/g of creatinine (10 mg/g) excreted [37]. Levels of urine albumin excretion exceeding this threshold of 10 mg/g are associated with increased risk of CVD, end-stage kidney disease, and mortality [38]. Increased urine albumin excretion is also hypothesized to reflect endothelial dysfunction, a central feature in metabolic syndrome. To understand the ties between insulin resistance, endothelial dysfunction, and increased urine albumin excretion, it is important to remember the key roles of insulin at the cellular level as shown in Fig. 39.1.

Fig. 39.1 demonstrates the normal function of insulin, the insulin receptor, and its effects on an endothelial cell. As shown in Fig. 39.1, insulin binds to an insulin receptor on the cell membrane. The insulin receptor is a transmembrane receptor that includes two extracellular alpha subunits and two transmembrane beta subunits linked by sulfide bonds [39]. Binding of insulin to the alpha subunits of the insulin receptor induces a conformational change that turns on kinase activity in beta subunits. This then leads to phosphorylation of the beta subunits on tyrosine residues and further activation of the intracellular tyrosine kinase domain. This phosphorylation is critically important for initiating the actions of the insulin itself [40]. Tyrosine phosphorylation of an insulin receptor via binding with insulin activates phosphoinositide 3-kinase (PI3 kinase) that leads to phosphorylation of phosphatidylinositol 4,5 bisphosphate (PIP2) to yield phosphatidylinositol 3,4,5-triphosphate or PIP3. PIP3 is important for recruitment and activation of protein kinase B known as Akt, a serine, threonine kinase that regulates multiple cellular functions. Once Akt is activated via phosphorylation via the mammalian target of rapamycin complex 2, it recruits and stimulates several messengers to yield the main actions of insulin within the cell. For example, Akt leads to insertion of Glut4 transporters into the cell membrane wall—allowing glucose to enter the muscle cell [41]. Akt also phosphorylates and inactivates glycogen synthase kinase 3 that stimulates synthesis of glycogen from glucose and inhibits glycogeneolysis [42]. Activation of the mammalian target of rapamycin complex 1 by Akt leads to phosphorylation and inhibition of proteins kinases that

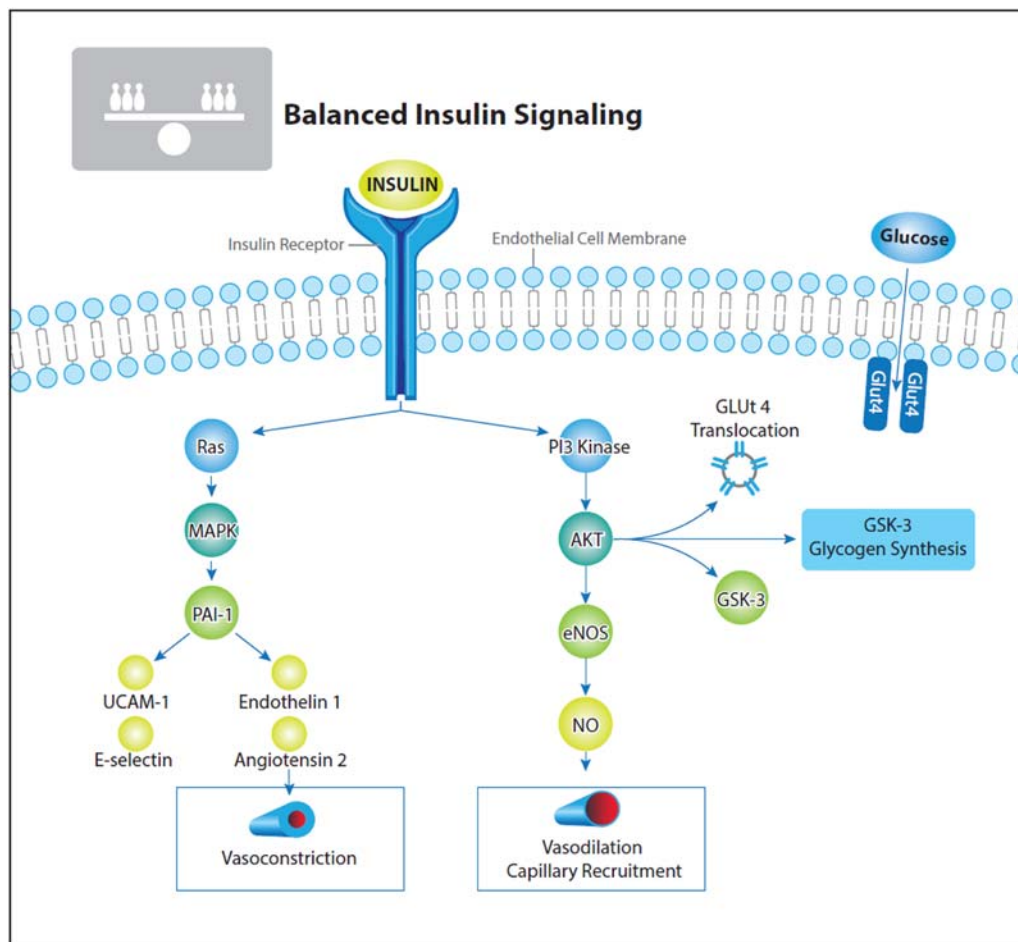


FIGURE 39.1 Figure depicts balanced insulin signaling in a healthy state. Insulin binds to insulin receptor and leads to activation of several G protein kinases and activation of both MAPK and AKT. MAPK activation leads to upregulation of vasoconstrictor production, for example, UCAM-1, endothelin-1, and angiotensin-2. Simultaneously, AKT activation leads to upregulation of endothelial nitric oxide synthase activity, increasing NO production. *AKT*, Protein kinase B; *eNOS*, endothelial nitric oxide synthase; *GSK-3*, glycogen synthase 3 kinase; *GLUT4*, glucose transporter type 4; *MAPK*, mitogen-activated protein kinase; *NO*, nitric oxide; *PAI-1*, plasminogen activator inhibitor-1; *PI3* kinase, phosphoinositide 3-kinase; *UCAM-1*, intracellular adhesion molecule-1.

regulate expression of genes controlling protein synthesis and cell growth [39]. The phosphorylation and translocation from nucleus to cytoplasm of the transcription factors of the Forkhead Box O1 (FOXO1) by Akt blocks FOXO1 transcriptional activity. FOXO1 accelerates autophagy in response to cellular stress.

Thus the actions of insulin are not singularly focused on glucose transport but rather multiplex. Insulin is important for capillary recruitment and vasodilation to enhance cellular uptake of glucose and this occurs via the actions of endothelial nitric oxide synthase (eNOS). Activated Akt phosphorylates and activates eNOS. This is balanced by the actions of insulin on mitogen-activated protein (MAP) kinase. The activation of MAP kinase by tyrosine phosphorylation leads to upregulation of endothelin 1 production and expression of vascular cell adhesion molecule VCAM-1 and E-selectin-inducing more leukocyte endothelial interactions, endothelial cell growth,

and vasoconstriction. In healthy states, mediators of vasoconstriction and cellular proliferation are balanced with mediators of vasodilation and inhibitors of cellular proliferation (Fig. 39.1). In the setting of insulin resistance, the MAP kinase arm is not affected and can even be upregulated. In contrast, the PI3-Akt pathway is downregulated that leads to lower eNOS activation and less production of NO, a vasodilator (Fig. 39.2). In addition, the lack of cellular glucose uptake via Glut4 means that more glucose can enter the hexosamine biosynthetic pathway that normally metabolizes less than 3% of all glucose. The major end product of this pathway is uridine diphosphate N-acetylglucosamine (UDP-GlcNAc) and elevated levels of this can lead to modification of eNOS with O-GlcNAc known as O-GlcNAcylation [43] that inactivates eNOS leading to reduced levels of NO. Elevated levels of UDP-GlcNAc have been implicated in initiating and sustaining insulin resistance but this remains controversial.

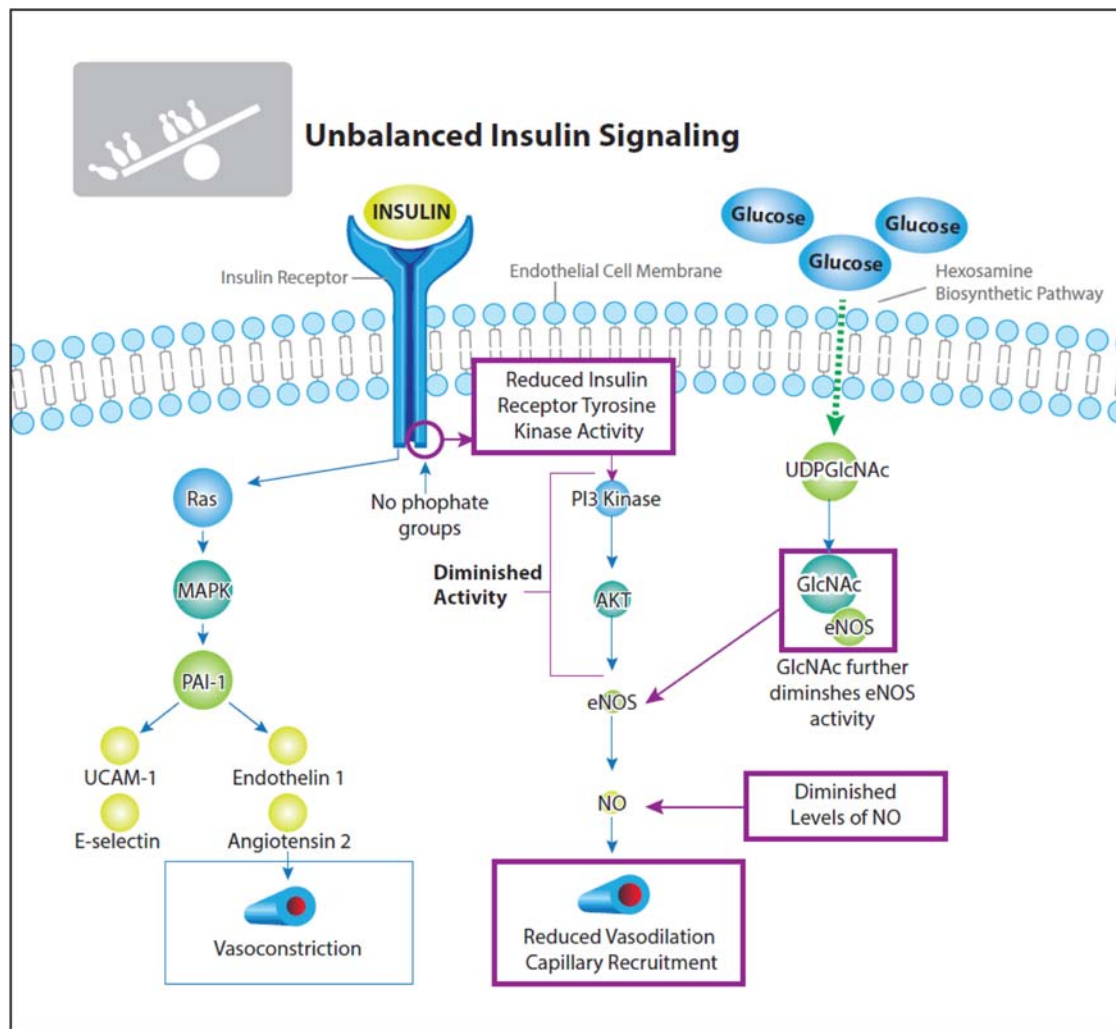


FIGURE 39.2 Unbalanced insulin signaling in an insulin-resistant state. Insulin binds to insulin receptor but due to resistance, G protein kinases are less active. MAPK is not affected but AKT is not less phosphorylated and less active leading to lower GLUT4 translocation to cell membrane and less glucose entering the cell. This leads to more glucose entering the hexosamine biosynthetic pathway. The major endproduct of hexosamine biosynthetic pathway is UDP-GlcNAc that modifies eNOS making it less active. Activation of AKT is also important for activating eNOS. The end result is less NO while vasoconstrictor production is not affected. AKT, Protein kinase B; eNOS, endothelial nitric oxide synthase; GlcNAc, N-acetylglucosamine; GLUT4, glucose transporter type 4; GSK-3, glycogen synthase 3 kinase; MAPK, mitogen-activated protein kinase; NO, nitric oxide; PAI-1, plasminogen activator inhibitor-1; PI3 kinase, phosphoinositide 3-kinase; UCAM-1, intracellular adhesion molecule-1; UDP-GlcNAc, uridine diphosphate N-acetylglucosamine.

The imbalance between vasoconstrictors (endothelin) versus vasodilators (NO) creates a vicious cycle because the vasoconstriction inhibits glucose uptake due to lower capillary recruitment and decreased local blood flow that is normally mediated via the actions of insulin [44,45]. To interrupt this cycle, eNOS activity needs to be upregulated; eNOS phosphorylation by Akt kinase can be heightened with statin medications and by adiponectin and via interruption of insulin resistance via reduction in caloric intake and weight loss (Table 39.2). Thus insulin resistance is marked by an imbalance of vasoconstrictors, vascular cell adhesion, and proliferation versus vasodilators. This leads to endothelial dysfunction in endothelial cells.

The actions of insulin resistance can also be seen within podocytes that also have insulin receptors. High extracellular glucose can induce podocyte apoptosis and both insulin resistance and glucose toxicity on podocyte function and autophagy have been implicated in the etiology of diabetic nephropathy [46,47]. Cross-sectional studies have demonstrated an association between increased urine albumin excretion and visceral adiposity, especially in adults with type 2 diabetes [48–50]. However, this association between visceral adiposity and increased urine albumin excretion may be due to the effects of elevated BP alone, and not a result of metabolic syndrome per se [51,52].

TABLE 39.2 Factors that influence endothelial nitric oxide synthase (eNOS) phosphorylation by Akt activity.

Factor	Mechanism
Weight loss/Caloric restriction	Interrupts insulin resistance via reducing glucose entering hexosamine biosynthetic pathway and reducing production of uridine diphosphate N-acetylglucosamine, an inhibitor of eNOS and via upregulation of adiponectin production
Adiponectin	Adiponectin phosphorylates eNOS at Ser ¹¹⁷⁷ and activates AMP-activated protein kinase
Statin medications	Blocks formation of many compounds, such as farnesylpyrophosphate and geranylgeranyl pyrophosphate, nonsteroidal isoprenoid compounds, that inactivate G-proteins important for upregulating eNOS activity

The link between obesity and kidney disease is not limited to increased urine albumin excretion. Obesity, especially morbid obesity, strongly impacts risk of all stages of CKD. Obesity, especially morbid obesity, appears to be a very strong risk factor for future risk of end-stage renal disease but this association appears strongest if obesity occurs during the early years of adulthood. One of the largest cohort studies in the United States examined 320, 252 adults who were age 18–34 years when they participated in a health screening exam during years 1964 and 1985. After 15–34 years of follow-up, morbid obesity (BMI ≥ 40 kg/m²) was associated with a sixfold higher risk of kidney failure compared to an ideal BMI (18.5–24.9 kg/m²) [53]. A cohort that followed adolescents thru adulthood also found similar associations of morbid obesity with kidney failure [54]. In contrast, obesity occurring later in life has not demonstrated strong associations between obesity and kidney failure risk [55–57]. The age modification of kidney disease risk associated with obesity may be due to cumulative exposure to high insulin and glucose levels if obesity begins at an early age. This hypothesis is supported by studies showing that associations of obesity with kidney failure risk is limited to individuals with metabolic syndrome and/or elevated BP [55,58]. In the REasons for Geographic and Racial Differences in Stroke (REGARDS) study, metabolic syndrome was a strong modifier of the association of obesity with kidney disease. Among 21,840 REGARDS participants with a mean baseline age of 65 years and followed for 6.3 ± 1.3 years, 247 developed kidney failure. Obesity defined by a BMI ≥ 30 kg/m² was associated with a twofold higher risk for kidney failure compared to an ideal BMI of 18.5–24.9 kg/m² after adjustment for covariates. However, this association of higher kidney failure risk with obesity status was only noted in the REGARDS participants with metabolic syndrome [55]. Similar findings were also noted in a large cohort of adults followed for risk of incident CKD [59]. Measures of obesity were not associated with incident CKD in the absence of metabolic syndrome traits.

Measures of abdominal obesity such as increased waist circumference or waist-to-hip ratio are more consistently associated with kidney disease risk compared to BMI [56,60,61]. In addition, multiple studies have consistently demonstrated an association of abdominal obesity and increased urine protein (mainly albumin) excretion in several types of populations [48,49,62]. Differences in kidney disease risk across obesity measures are likely due to the fact that waist circumference and waist-to-hip ratio reflect visceral fat, a strong risk factor for development of insulin resistance, while BMI includes visceral fat and lower body fat. Even with these visceral adiposity measures, obesity among older adults (e.g., 60 years and older) only modestly increases the risk of kidney failure and associations appear largely driven by hypertension and diabetes. In contrast, if obesity, especially morbid obesity, initiates during adolescence or young adulthood then the individual is exposed to metabolic syndrome traits for a prolonged period of time, heightening kidney disease risk.

Metabolic syndrome defined by the NCEP ATP III criteria is associated with moderately increased odds of both incident and prevalent CKD [63–65]. Odds of CKD defined as an estimated glomerular filtration rate (eGFR) < 60 mL/min/m² or a spot urine albumin/creatinine ratio (30–300 mg/g) increased linearly with the number of metabolic syndrome traits from 2.21 [95% confidence interval (CI), 1.16–4.24] with 2 traits to 5.85 (95% CI, 3.11–11.0) with 5 traits compared to individuals with no traits. In a cross-sectional analysis of a population-based survey of the US population [64], similar findings were noted in the Jackson Heart Study, a cohort of African American adults. Among the Jackson Heart cohort, elevated BP, triglycerides, fasting blood glucose, and abdominal obesity were all significantly associated with increased odds of CKD [66]. However, the presence of metabolic syndrome was associated with a 2.2-fold increased odds of CKD (95% CI, 1.78–2.78). Among persons with established CKD such as the African-American Study of Hypertension and Kidney Disease, in which all trial participants had CKD at baseline, the

presence of metabolic syndrome was associated with higher levels of urine protein excretion at baseline but not with CKD progression [67]. In the Strong Heart Study a cohort of American Indian adults demonstrated that the presence of metabolic syndrome was associated with heightened risk of incident CKD but this association was largely driven by development of diabetes and hypertension [68]. Most studies find moderate associations of metabolic syndrome with incident CKD. A meta-analysis published in 2011 found eleven studies with a total of 30,146 adults followed over time for CKD defined as an eGFR < 60 mL/min/1.73 m². The presence of metabolic syndrome was associated with a 1.6-fold increased odds of CKD (95% CI, 1.3, 1.8) and the higher the number of metabolic syndrome traits present, the stronger the association (trend *P* value = .02). The strongest association with CKD for a given metabolic syndrome trait was noted for elevated BP with an odds ratio (OR) of 1.6 (95% CI 1.3, 2.0) and lowest for abdominal obesity (OR 1.2; 95% CI 1.1, 1.3), and impaired fasting glucose (OR 1.1; 95% CI 1.0, 1.3). The metaanalysis also reported that three studies showed that metabolic syndrome was associated with an increased risk for development of increased urine albumin excretion [69].

Information on the link between metabolic syndrome and CKD risk remains fairly limited because most cases of kidney disease evolve over decades and not within a few years so cohort studies need extended follow up to fully evaluate potential risk factors. The metabolic syndrome trait most consistently and strongly associated with loss of eGFR over time in cohort studies is elevated BP. The Atherosclerosis Risk in Communities Study with 30 years of follow-up has demonstrated that the presence of hypertension is associated with a faster rate of eGFR decline and higher risk of CKD even after adjustment for diabetes status [70]. We also know that higher BP among adults with diabetes is associated with a higher risk of decreased GFR, especially in the presence of increased urine albumin excretion [71,72]. Based on metaanalyses of well-designed clinical trials, a systolic BP (SBP) target <130 mmHg is now recommended for individuals with nondialysis dependent CKD [73,74]. The evidence supporting this BP target is strongest for persons without diabetes mellitus [75] or persons with moderate to severely increased urine albumin excretion [72,73]. Among individuals with diabetes, intensive glucose control lowers risk of developing increased urine albumin excretion, especially during the early stages of diabetes. The Action to Control Cardiovascular Risk [76] in Diabetes study showed small but significant reductions in the absolute risk of developing increased urine albumin excretion with intensive glucose lowering. A metaanalysis of intensive glucose control studies in type 2 diabetes mellitus (T2DM) 1 [77] with 28,065

adults with type 2 diabetes followed for 2–15 years showed significantly reduced risk of both moderately increased [relative risk (RR) = 0.86, 95% CI: 0.76–0.96] and severely increased urine albumin excretion (RR = 0.74, 95% CI: 0.65–0.85) with intensive glucose control [median hemoglobin A1c (HbA1c) 6.4%–7.4%]. To date, intensive glucose control has not been associated with reduced risk of kidney failure although data suggest trends toward benefits with intensive control [78]. Several studies have also shown that even among adults without established diabetes, higher HbA1c values are associated with faster rate of age-associated decline in the eGFR [79,80]. In a secondary analysis of the Outcome Reduction with Initial Glargine Intervention trial that included 12,537 people with T2DM or prediabetes followed for 6.2 years, RR of kidney failure was 54% higher (95% CI; 24%, 91%) for every 1% higher baseline HbA1c values among trial participants. In addition, long-term glucose control with pancreas transplant had demonstrated reversibility of diabetic nephropathy posttransplant [81], but this was only seen histologically after 10 years of normal glucose levels [82].

Higher urine albumin or protein excretion is associated with faster GFR decline in the setting of kidney disease [83–85]. While BP-lowering will lower urine protein excretion [71,86], caloric restriction and weight loss are also effective interventions regardless of BP because weight loss improves insulin resistance via downregulation of fatty acid mobilization [87] and reductions in inflammation and renin–angiotensin–aldosterone system activation [88]. High concentration of circulating free fatty acids (FFAs) promotes insulin resistance by competitively inhibiting glucose uptake by muscle. For example, urine protein excretion decreases rapidly after bariatric surgery among morbidly obese adults likely because caloric intake during the first 6 months is markedly reduced [89]. Declines in urine protein excretion occur as early as 1 month with caloric restriction. A metaanalysis of 13 studies of weight-loss interventions, including dietary restriction, exercise, antiobesity medications, and surgery with a total of 522 participants [90] showed that urine protein or albumin excretion decreased significantly with weight loss, regardless of the intervention. On average, urine protein excretion decreased by 110 mg (95% CI, 60–160 mg, *P* < .001) and urine albumin excretion decreased by 1.1 mg (95% CI, 0.5–2.4 mg, *P* = .011) per kilogram of weight loss, for both surgical and medical interventions.

One cellular metabolic pathway that elucidates mechanisms whereby caloric restriction reduces protein excretion is upregulation of adiponectin that improves insulin sensitivity at the cellular level. Adiponectin receptors reside on podocytes and may improve function and survival [91,92]. Adiponectin

expression is upregulated via the Sirt1-Foxo1 pathway [93]. Sirt1, an NAD⁺-dependent deacetylase, expressed in the renal inner medulla, activates the cell cycle protein Foxo1 [93,94], which then forms a transcriptional complex with C/EBP α at the adiponectin gene promoter site and turns on gene expression. Adiponectin can upregulate eNOS activity via phosphorylation of eNOS at Serine residue 1177 similar to actions of Akt previously described [95]. Increased activity of eNOS leads to higher NO levels and increased blood flow and vasodilation and can reduce BP.

Treatment of metabolic syndrome

Because metabolic syndrome confers such a strong risk for cardiovascular and kidney diseases and future development of type 2 diabetes, the syndrome should be addressed with a diet and lifestyle intervention. The best evidence we have for amelioration of metabolic syndrome with a lifestyle intervention is from the Diabetes Prevention Program (DPP). The DPP [96] randomized persons at high risk for type 2 diabetes to either an intensive lifestyle intervention, metformin, or a placebo. After 3 years, persons in the intensive lifestyle intervention had a 58% lower risk of developing type 2 diabetes compared to the placebo group. The annual risk of developing type 2 diabetes was 5% in the lifestyle intervention group and 11% in the placebo group. The benefits of lifestyle changes were most prominent among persons age 60 years and older in whom risk of developing type 2 diabetes was reduced by 71% compared to placebo. Metformin use was also associated with reduced risk of developing type 2 diabetes compared to placebo. Risk of developing type 2 diabetes was 31% lower compared to placebo and effects were strongest among women with a history of gestational diabetes and among those with a baseline BMI of 35 kg/m² and higher. After the trial ended, participants were then followed for the development of type 2 diabetes. After 15 years of follow-up, 55% and 56% of participants assigned to the lifestyle intervention and metformin, respectively, developed type 2 diabetes compared to 62% of participants assigned to placebo during the trial [97]. Similar findings have been shown in similar trials completed outside the United States in which diet and lifestyle changes led to approximately 30%–40% lower risk of type 2 diabetes over 10–20 years [98,99].

Lifestyle interventions for metabolic syndrome must include both weight loss and physical activity. In the DPP trial, weight loss goals were 7% or higher of baseline weight and physical activity goals were set at 150 minutes of vigorous activity such as brisk walking to expend approximately 700 kcal/week. Weight loss

recommendations were 1–2 lbs/week with reducing between 500 and 1000 cal/day depending on the baseline body weight.

The type of diet may also help reduce risk of type 2 diabetes as implementation of a Mediterranean diet high in monosaturated fats such as olive oil and low in saturated fats has been shown to reduce risk of type 2 diabetes and kidney disease [100]. In addition, observational studies suggest that persons with high adherence to a Mediterranean diet have a lower risk of developing metabolic syndrome [101]. The Western diet is high in processed foods, sugar, and fat and these factors all increase risk of developing metabolic syndrome and T2DM.

Metformin was approved for the prevention of type 2 diabetes by the FDA in 2017 and is recommended by the American Diabetes Association for diabetes prevention in high-risk individuals [102]. Metformin is a biguanide that inhibits hepatic glucose output. However, only 50% of the drug is absorbed and reaches the liver while the other 50% accumulates in the gut. Within the gut, metformin alters the microbiome by increasing longevity of gut bacteria that consume calories within the intestine [103].

Other medications that may be considered for addressing metabolic syndrome and insulin resistance are the thiazolidinediones (TZDs). The TZD class of glucose-lowering medications promotes insulin sensitivity by activating the peroxisome proliferator-activated receptor-gamma that leads to upregulation of genes involved with lipid metabolism and adipogenesis. The use of this class of drugs is associated with reduction in FFAs and urine albumin excretion in adults with type 2 diabetes via amelioration of insulin sensitivity in liver, muscle, and fat tissue [104]. Trials of TZDs have not shown that this class of medications slow kidney function decline or reduce risk of incident CKD or kidney failure. The use of this medication has been impeded by its risks of sodium retention and weight gain leading to heart failure.

Another class of drugs, the glucagon-like peptide-1 (GLP-1) receptor agonists, may become increasingly more relevant for management of metabolic syndrome [105]. GLP-1, a peptide hormone, is secreted in response to circulating glucose and increases pancreatic insulin secretion and decreases glucagon secretion. Normally, GLP-1 is rapidly degraded by the enzyme dipeptidyl peptidase IV but the GLP-1 receptor agonists are resistant to degradation by this enzyme. Receptors for GLP-1 reside in multiple tissues in the body, including the hypothalamus, pancreas, and kidney. Currently, drugs in this class are approved for use in persons with type 2 diabetes but one drug, liraglutide, is approved for management of obesity by the Federal Drug Administration. These drugs bind to

receptors in the hypothalamus and increase satiety and decrease appetite. Liraglutide use is generally associated with weight loss (~6 kg) and other GLP-1 agonists can also lead to weight loss, albeit less. This drug class is also associated with an approximate 5 mmHg drop in SBP and diastolic BP and a reduction in serum triglycerides. Effects on metabolic syndrome traits are likely mediated mainly via weight loss.

While this drug class is not approved for treatment of metabolic syndrome, metabolic syndrome traits may be positively affected with GLP-1 agonists. Animal models that lack GLP-1 receptors show higher urinary albumin levels and mesangial expansion in the setting of diabetes and this is attributed to higher levels of oxidative stress due to increased glomerular superoxide. Liraglutide in animals lacking GLP-1 receptors ameliorates the effects of hyperglycemia in the kidney by suppressing superoxide production. In the Liraglutide Effect and Action in Diabetes: Evaluation of Cardiovascular Outcome Results trial, the use of liraglutide was associated with a 22% lower risk of a primary kidney endpoint defined as the development of macroalbuminuria, doubling of serum creatinine and $\text{eGFR} < 45 \text{ mL/min/1.73 m}^2$, the need for renal replacement therapy, and death from renal causes (95% CI: 8%, 33%; $P = .03$) compared to placebo. However, most of the renal endpoints in this trial were new-onset macroalbuminuria. The renal benefit of liraglutide was mainly derived from a 26% significant reduction in new-onset macroalbuminuria (95% CI 9%, 40%) without any significant changes in eGFR . The use of liraglutide in this trial was associated with a 2.3 kg weight loss and a drop of 1.2 mmHg in SBP. Similar findings were noted in the Trial to Evaluate Cardiovascular and other long-term outcomes with semaglutide (SUSTAIN-6) with 46% reduction (95% CI 23%, 63%; $P < .001$) in risk of developing macroalbuminuria relative to placebo among adults with type 2 diabetes [106].

Another drug class with demonstrated benefits on metabolic syndrome traits is the dipeptidyl peptidase-4 inhibitors. This class of drugs increases beta cell mass and function and improves liver, muscle, and adipose tissue insulin sensitivity [107]. While fatty acid oxidation is increased with DPP-4 inhibitors, hepatic triglyceride synthase is reduced leading to lower triglyceride levels and decreased lipogenesis. These drugs also delay gastric emptying potentiating satiety leading to decreased caloric intake. A pooled analysis of 13 clinical trials of the DPP-4 inhibitor linagliptin versus placebo [108] with follow-up ranging from 12 to 76 weeks showed linagliptin reduced risk of developing a kidney endpoint defined as new-onset micro or macroalbuminuria, loss of baseline $\text{eGFR} > 50\%$, acute renal failure, or death from any cause. The absolute risk difference in kidney endpoints was 2.8% with lower risk in the linagliptin group (HR = 0.84; 95%

CI: 0.72–0.97, $P = .02$). Currently, the DPP-4 inhibitors are only approved for treatment of type 2 diabetes.

Another class of drugs to consider for management of metabolic syndrome in certain patients is the sodium–glucose cotransporter type 2 (SGLT2) receptor inhibitors, especially for patients with a high risk of kidney failure. One of these drugs, canagliflozin, was the first diabetes drug approved by the Federal Drug Administration for the prevention of cardiovascular events. The SGLT2 receptor is found in the early proximal tubule and is responsible for mediating sodium and glucose reabsorption. When SGLT2 is inhibited, glucose and sodium are excreted in the urine. Due to the loss of sodium, which is sensed by the macula densa in the juxtaglomerular apparatus, intraglomerular pressure is reduced due to afferent arteriolar vasoconstriction. The loss of glucose leads to lower insulin levels and these effects along with sodium loss explain the diuretic effects of this drug class. The diuresis leads to weight loss and lowered BP. In the CVD safety trial of empagliflozin (EMPA-REG), use of empagliflozin in 7020 adults with type 2 diabetes and high CVD risk showed significantly lower risk of kidney outcomes defined as progression to macroalbuminuria, doubling of serum creatinine level associated with an $\text{eGFR} < 45 \text{ mL/min/1.73 m}^2$, initiation of renal replacement therapy and death from renal disease. Absolute risk reduction of renal endpoints with empagliflozin was 6.1%—with events occurring in 12.7% of those treated with empagliflozin and 18.8% in the placebo group ($P < .001$). Similar to other trials of glucose-lowering medications, most renal endpoints were increased urine albumin excretion, and in the EMPA-REG trial, reduction in new-onset macroalbuminuria was responsible for most of the renal benefits with empagliflozin. In animal models, this drug class shows reduction in oxidative stress, inflammation, fibrosis, and tubular senescence that would provide direct kidney benefits [109]. A metaanalysis of four trials of SGLT2 inhibitors in 38,723 adults with type 2 diabetes [110], SGLT2 inhibitors reduced the risk of dialysis, transplantation, or death due to kidney disease by 33% (95% CI 14%, 48%, $P = .0019$) relative to placebo. Risks of both kidney failure (RR = 0.65, 0.53–0.81, $P < .0001$) and acute kidney injury (RR = 0.75, 0.66–0.85, $P < .0001$) were significantly reduced with SGLT2 inhibitor use. Benefits were consistent across baseline urine albumin excretion levels and use of renin–angiotensin system inhibitors. Because these drugs do not increase insulin secretion, they may have benefits in adults without established diabetes and could reduce kidney disease risk in the setting of metabolic syndrome.

The CREDENCE Trial [111] examined canagliflozin in adults with type 2 diabetes and kidney disease, defined as an $\text{eGFR} 30\text{--}90 \text{ mL/min/1.73 m}^2$ and

albumin-to-creatinine ratio 300–5000 mg/g and receiving treatment with a renin–angiotensin system blocker. The trial was stopped early after an interim analysis showed the risk of kidney failure and cardiovascular events was 30% (95% CI, 18%, 41%; $P = .00001$) lower in the canagliflozin group than in the placebo group at a median follow-up of 2.62 years. Kidney endpoints were defined as development of kidney failure, doubling of baseline serum creatinine level, or death from renal causes. Risk of the composite renal outcome was 34% (95% CI, 18%, 47%; $P < .001$) relative to placebo while risk of kidney failure was 32% lower (95% CI, 14%, 46%; $P = .002$) relative to placebo. Risk of cardiovascular events defined as cardiovascular death, myocardial infarction, or stroke was reduced by 20% (95% CI, 5%, 33%; $P = .01$) while hospitalization for heart failure was reduced by 39% (95% CI, 20%, 53%; $P < .001$) relative to placebo. In contrast to the CREDENCE Trial, the study to evaluate the effects of dapagliflozin on renal outcomes and cardiovascular mortality in patients with CKD NCT03036150 is enrolling patients with both diabetic and nondiabetic CKD and will help determine whether this drug class shows similar benefits to adults with type 2 diabetes and may provide evidence for its use in metabolic syndrome.

Finally, the importance of lifestyle modifications, including dietary changes, cannot be overstated for those with metabolic syndrome and obesity. Diets that emphasize the consumption of whole, plant-based foods have been repeatedly linked with weight loss, likely due to the presence of fiber in plant-based foods and other differences [112]. Several observational studies have shown that increasing fiber consumption is directly associated with weight loss [113]. For example, in one study, every 1-g increase in dietary fiber intake was associated with a 0.25-kg reduction in body weight [114]. A systematic review and metaanalysis of clinical trials found that vegetarian diets were associated with a 3.4-kg weight loss in an intention-to-treat analysis, with larger weight losses being achieved in those who completed the trials [115]. Fiber's ability to facilitate weight loss is likely related to the lower caloric density of those foods and increases in satiety hormones like GLP-1 and peptide YY (both of which are increased by soluble fiber) [116].

In addition to weight loss, the consumption of plant-based foods may also have benefits for kidney-related outcomes. In the Adventist Health Study-2, researchers found that among nearly 73,000 participants followed over an average of 5.8 years, those consuming a vegetarian diet had a 52% reduced risk (95% CI, 8%, 82%) of kidney disease associated mortality [117]. Consumptions of diets emphasizing plant-based foods have also been associated with less proteinuria and a lower risk of CKD [118,119]. Benefits of plant-based foods may also occur with only partial inclusion of these foods into a diet. In a

randomized, controlled trial of the use of fruits and vegetables for metabolic acidosis secondary to CKD, Goraya et al. found reductions in BP, weight, and albuminuria along with improvements in metabolic acidosis comparable to oral bicarbonate with as little as two to four servings of fruits and vegetables per day [120]. Other benefits of a plant-based diet in kidney disease may include reductions in serum phosphate levels and uremic toxins [121].

In summary, metabolic syndrome is defined by the presence of multiple traits that occur due to caloric consumption exceeding energy expenditure. Persons with a genetic predisposition to store these excess calories in the abdomen and those with reduced fat storage capacity have increased susceptibility to develop metabolic syndrome. Metabolic syndrome is associated with increased risk of CKD. Weight loss, dietary changes, and exercise remain the cornerstone treatment for metabolic syndrome. Medications that improve insulin sensitivity or impair kidney glucose reabsorption may also help ameliorate the impact of metabolic syndrome on kidney disease risk.

References

1. NEEL JV. Diabetes mellitus: a "thrifty" genotype rendered detrimental by "progress"? *Am J Hum Genet* 1962;14:353–62.
2. Hales CM, Carroll MD, Fryar CD, Ogden CL. Prevalence of obesity among adults and youth: United States, 2015–2016. *NCHS Data Brief* 2017;(288):1–8 288.
3. World Health Organization. Fact sheet. Obesity and overweight. 2018.
4. Herrera BM, Lindgren CM. The genetics of obesity. *Curr Diab Rep* 2010;10(6):498–505. Available from: <https://doi.org/10.1007/s11892-010-0153-z>.
5. Lee MJ, Wu Y, Fried SK. Adipose tissue heterogeneity: implication of depot differences in adipose tissue for obesity complications. *Mol Asp Med* 2013;34(1):1–11. Available from: <https://doi.org/10.1016/j.mam.2012.10.001>.
6. Rask-Andersen M, Karlsson T, Ek WE, Johansson A. Genome-wide association study of body fat distribution identifies adiposity loci and sex-specific genetic effects. *Nat Commun* 2019;10(1):339–018–08000–4. Available from: <https://doi.org/10.1038/s41467-018-08000-4>.
7. Spalding KL, Arner E, Westermark PO, et al. Dynamics of fat cell turnover in humans. *Nature* 2008;453(7196):783–7. Available from: <https://doi.org/10.1038/nature06902>.
8. Reaven GM. Banting lecture 1988. Role insulinresistance human disease. *Diabetes* 1988;37(12):1595–607.
9. Grundy SM, Brewer Jr HB, Cleeman Jr, et al. Definition of metabolic syndrome: report of the National Heart, Lung, and Blood Institute/American Heart Association conference on scientific issues related to definition. *Circulation* 2004;109(3):433–8. Available from: <https://doi.org/10.1161/01.CIR.000011245.75752.C6>.
10. World Health Organization. Obesity: preventing and managing the global epidemic: report of a WHO consultation on obesity, Geneva, June 3–5, 1997. 1998.
11. Kaplan NM. The deadly quartet. Upper-body obesity, glucose intolerance, hypertriglyceridemia, and hypertension. *Arch Intern Med* 1989;149(7):1514–20. Available from: <https://doi.org/10.1001/archinte.149.7.1514>.

12. National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III). Third Report of the National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III) final report. *Circulation* 2002;106(25):3143–421.
13. Balkau B, Deanfield JE, Despres JP, et al. International Day for the Evaluation of Abdominal Obesity (IDEA): a study of waist circumference, cardiovascular disease, and diabetes mellitus in 168,000 primary care patients in 63 countries. *Circulation* 2007;116(17):1942–51. Available from: <https://doi.org/10.1161/CIRCULATIONAHA.106.676379>.
14. Zimmet P, Magliano D, Matsuzawa Y, Alberti G, Shaw J. The metabolic syndrome: a global public health problem and a new definition. *J Atheroscler Thromb* 2005;12(6):295–300 doi: JST.JSTAGE/jat/12.295 [pii].
15. Kylin E. [Studies of the hypertension-hyperglycemia-hyperuricemia syndrome] Studien uber das hypertonic-hyprglykamic-hyperurikamies syndrome. *Zentralblatt fur Innere Med* 1923;44:105–7.
16. Herman JB, Medalie JH, Goldbourt U. Diabetes, prediabetes and uricaemia. *Diabetologia* 1976;12(1):47–52. Available from: <https://doi.org/10.1007/bf01221964>.
17. Alberti KG, Eckel RH, Grundy SM, et al. Harmonizing the metabolic syndrome: a joint interim statement of the International Diabetes Federation Task Force on Epidemiology and Prevention; National Heart, Lung, and Blood Institute; American Heart Association; World Heart Federation; International Atherosclerosis Society; and International Association for the Study of Obesity. *Circulation* 2009;120(16):1640–5. Available from: <https://doi.org/10.1161/CIRCULATIONAHA.109.192644>.
18. Appel SJ, Oster RA, Floyd NA, Ovalle F. Cardiometabolic risk among African American women: a pilot study. *J Cardiovasc Nurs* 2009;24(2):140–50.
19. Liao Y, Kwon S, Shaughnessy S, et al. Critical evaluation of adult treatment panel III criteria in identifying insulin resistance with dyslipidemia. *Diabetes Care* 2004;27(4):978–83.
20. Cheal KL, Abbasi F, Lamendola C, McLaughlin T, Reaven GM, Ford ES. Relationship to insulin resistance of the adult treatment panel III diagnostic criteria for identification of the metabolic syndrome. *Diabetes* 2004;53(5):1195–200.
21. United States Department of Health and Human Services. The surgeon general's vision for a healthy and fit nation. 2010.
22. Brietzke SA. Controversy in diagnosis and management of the metabolic syndrome. *Med Clin North Am* 2007;91(6):1041–61 vii–viii. doi: S0025-7125(07)00078-8 [pii].
23. Reaven GM. Relationships among insulin resistance, type 2 diabetes, essential hypertension, and cardiovascular disease: similarities and differences. *J Clin Hypertens (Greenwich)* 2011;13(4):238–43. Available from: <https://doi.org/10.1111/j.1751-7176.2011.00439.x>.
24. Salazar MR, Carbajal HA, Espeche WG, et al. Relationships among insulin resistance, obesity, diagnosis of the metabolic syndrome and cardio-metabolic risk. *Diab Vasc Dis Res* 2011;8(2):109–16. Available from: <https://doi.org/10.1177/1479164111403170>.
25. Moore JX, Chaudhary N, Akinyemiju T. Metabolic syndrome prevalence by race/ethnicity and sex in the United States, National Health and Nutrition Examination Survey, 1988–2012. *Prev Chronic Dis* 2017;14:E24. Available from: <https://doi.org/10.5888/pcd14.160287>.
26. Heiss G, Snyder ML, Teng Y, et al. Prevalence of metabolic syndrome among Hispanics/Latinos of diverse background: the Hispanic Community Health Study/Study of Latinos. *Diabetes Care* 2014;37(8):2391–9. Available from: <https://doi.org/10.2337/dc13-2505>.
27. Kahn SE, Prigeon RL, Schwartz RS, et al. Obesity, body fat distribution, insulin sensitivity and Islet beta-cell function as explanations for metabolic diversity. *J Nutr* 2001;131(2):354S–360SS. Available from: <https://doi.org/10.1093/jn/131.2.354S>.
28. Caldwell SH, Ikura Y, Iezzoni JC, Liu Z. Has natural selection in human populations produced two types of metabolic syndrome (with and without fatty liver)? *J Gastroenterol Hepatol* 2007;22(1):S11–19 Suppl.
29. Musso C, Cochran E, Moran SA, et al. Clinical course of genetic diseases of the insulin receptor (type A and Rabson–Mendenhall syndromes): a 30-year prospective. *Medicine* 2004;83(4):209–22.
30. Lara-Castro C, Garvey WT. Intracellular lipid accumulation in liver and muscle and the insulin resistance syndrome. *Endocrinol Metab Clin North Am* 2008;37(4):841–56.
31. Mattar SG, Velcu LM, Rabinovitz M, et al. Surgically-induced weight loss significantly improves nonalcoholic fatty liver disease and the metabolic syndrome. *Ann Surg* 2005;242(4):610–17.
32. Promrat K, Kleiner DE, Niemeier HM, et al. Randomized controlled trial testing the effects of weight loss on nonalcoholic steatohepatitis. *Hepatology* 2010;51(1):121–9.
33. Rafiq N, Younossi ZM. Effects of weight loss on nonalcoholic fatty liver disease. *Semin Liver Dis* 2008;28(4):427–33.
34. Goodpaster BH, Thaete FL, Simoneau JA, Kelley DE. Subcutaneous abdominal fat and thigh muscle composition predict insulin sensitivity independently of visceral fat. *Diabetes* 1997;46(10):1579–85.
35. Redon J, Rovira E, Miralles A, Julve R, Pascual JM. Factors related to the occurrence of microalbuminuria during antihypertensive treatment in essential hypertension. *Hypertension* 2002;39(3):794–8.
36. Meigs JB, D'Agostino Sr RB, Nathan DM, Rifai N, Wilson PW, Framingham Offspring S. Longitudinal association of glycemia and microalbuminuria: the Framingham Offspring Study. *Diabetes Care* 2002;25(6):977–83.
37. Mattix HJ, Hsu CY, Shaykevich S, Curhan G. Use of the albumin/creatinine ratio to detect microalbuminuria: implications of sex and race. *J Am Soc Nephrol* 2002;13(4):1034–9.
38. Chronic Kidney Disease Prognosis Consortium, Matsushita K, van der Velde M, et al. Association of estimated glomerular filtration rate and albuminuria with all-cause and cardiovascular mortality in general population cohorts: a collaborative meta-analysis. *Lancet* 2010;375(9731):2073–81.
39. Boucher J, Kleinridders A, Kahn CR. Insulin receptor signaling in normal and insulin-resistant states. *Cold Spring Harb Perspect Biol* 2014;6(1):a009191. Available from: <https://doi.org/10.1101/cshperspect.a009191> 10.1101/cshperspect.
40. Catalano KJ, Maddux BA, Szary J, Youngren JF, Goldfine ID, Schaufele F. Insulin resistance induced by hyperinsulinemia coincides with a persistent alteration at the insulin receptor tyrosine kinase domain. *PLoS One* 2014;9(9):e108693. Available from: <https://doi.org/10.1371/journal.pone.0108693>.
41. Miinea CP, Sano H, Kane S, et al. AS160, the Akt substrate regulating GLUT4 translocation, has a functional Rab GTPase-activating protein domain. *Biochem J* 2005;391(1):87–93 Pt doi: BJ20050887 [pii].
42. Hermida MA, Dinesh Kumar J, Leslie NR. GSK3 and its interactions with the PI3K/AKT/mTOR signalling network. *Adv Biol Regul* 2017;65:5–15 doi: S2212-4926(17)30124-0 [pii].
43. Laczky B, Hill BG, Wang K, et al. Protein O-GlcNAcylation: a new signaling paradigm for the cardiovascular system. *Am J Physiol Heart Circ Physiol* 2009;296(1):H13–28. Available from: <https://doi.org/10.1152/ajpheart.01056.2008>.
44. Keske MA, Premilovac D, Bradley EA, Dwyer RM, Richards SM, Rattigan S. Muscle microvascular blood flow responses in insulin resistance and ageing. *J Physiol* 2016;594(8):2223–31. Available from: <https://doi.org/10.1113/jphysiol.2014.283549>.

45. Muniyappa R, Sowers JR. Role of insulin resistance in endothelial dysfunction. *Rev Endocr Metab Disord* 2013;14(1):5–12. Available from: <https://doi.org/10.1007/s11154-012-9229-1>.
46. Coward RJ, Saleem MA. Podocytes as a target of insulin. *Curr Diabetes Rev* 2011;7(1):22–7 doi: BSP/CDR/E-Pub/00052 [pii].
47. Dai H, Liu Q, Liu B. Research progress on mechanism of podocyte depletion in diabetic nephropathy. *J Diabetes Res* 2017;2017:2615286. Available from: <https://doi.org/10.1155/2017/2615286>.
48. Kramer H, Reboussin D, Bertoni AG, et al. Obesity and albuminuria among adults with type 2 diabetes: the Look AHEAD (Action for Health in Diabetes) Study. *Diabetes Care* 2009;32(5):851–3.
49. Seo WJ, Lee GM, Hwang JH, Lee MN, Kang HC. Association between body mass index, waist circumference and prevalence of microalbuminuria in Korean adults of age 30 years and older without diabetes, hypertension, renal failure, or overt proteinuria: the 2013 Korean National Health and Nutrition Examination Survey. *Korean J Fam Med* 2016;37(1):57–63. Available from: <https://doi.org/10.4082/kjfm.2016.37.1.57>.
50. Burton JO, Gray LJ, Webb DR, et al. Association of anthropometric obesity measures with chronic kidney disease risk in a non-diabetic patient population. *Nephrol Dial Transplant* 2012;27(5):1860–6.
51. Knight EL, Kramer HM, Curhan GC. High-normal blood pressure and microalbuminuria. *Am J Kidney Dis* 2003;41(3):588–95. Available from: <https://doi.org/10.1053/ajkd.2003.50120>.
52. Griffin KA, Kramer H, Bidani AK. Adverse renal consequences of obesity. *Am J Physiol Ren Physiol* 2008;294(4):F685–96. Available from: <https://doi.org/10.1152/ajprenal.00324.2007>.
53. Hsu CY, McCulloch CE, Iribarren C, Darbinian J, Go AS. Body mass index and risk for end-stage renal disease. *Ann Intern Med* 2006;144(1):21–8.
54. Vivante A, Golan E, Tzur D, et al. Body mass index in 1.2 million adolescents and risk for end-stage renal disease. *Arch Intern Med* 2012;172(21):1644–50.
55. Panwar B, Hanks LJ, Tanner RM, et al. Obesity, metabolic health and the risk of end-stage renal disease. *Kidney Int* 2014;87(6):1216–22.
56. Kramer H, Gutierrez OM, Judd SE, et al. Waist circumference, body mass index, and ESRD in the REGARDS (Reasons for Geographic and Racial Differences in Stroke) Study. *Am J Kidney Dis* 2016;67(1):62–9. Available from: <https://doi.org/10.1053/j.ajkd.2015.05.023>.
57. Franceschini N, Gouskova NA, Reiner AP, et al. Adiposity patterns and the risk for ESRD in postmenopausal women. *Clin J Am Soc Nephrol* 2014;10(2):241–50. Available from: <https://doi.org/10.2215/CJN.0.02860314>.
58. Munkhaugen J, Lydersen S, Wideroe TE, Hallan S. Prehypertension, obesity, and risk of kidney disease: 20-year follow-up of the HUNT I study in Norway. *Am J Kidney Dis* 2009;54(4):638–46.
59. Chang AR, Surapaneni A, Kirchner HL, et al. Metabolically healthy obesity and risk of kidney function decline. *Obesity (Silver Spring)* 2018;26(4):762–8. Available from: <https://doi.org/10.1002/oby.22134>.
60. Elsayed EF, Sarnak MJ, Tighiouart H, et al. Waist-to-hip ratio, body mass index, and subsequent kidney disease and death. *Am J Kidney Dis* 2008;52(1):29–38. Available from: <https://doi.org/10.1053/j.ajkd.2008.02.363>.
61. Kovesdy CP, Czira ME, Rudas A, et al. Body mass index, waist circumference and mortality in kidney transplant recipients. *Am J Transpl* 2010;10(12):2644–51. Available from: <https://doi.org/10.1111/j.1600-6143.2010.03330.x>.
62. Lin WY, Pi-Sunyer FX, Liu CS, et al. Central obesity and albuminuria: both cross-sectional and longitudinal studies in Chinese. *PLoS One* 2012;7(12):e47960. Available from: <https://doi.org/10.1371/journal.pone.0047960>.
63. Chen B, Yang D, Chen Y, Xu W, Ye B, Ni Z. The prevalence of microalbuminuria and its relationships with the components of metabolic syndrome in the general population of China. *Clin Chim Acta* 2010;411(9–10):705–9.
64. Chen J, Muntner P, Hamm LL, et al. The metabolic syndrome and chronic kidney disease in United States adults. *Ann Intern Med* 2004;140(3):167–74.
65. Muntner P, Coresh J, Smith JC, Eckfeldt J, Klag MJ. Plasma lipids and risk of developing renal dysfunction: the atherosclerosis risk in communities study. *Kidney Int* 2000;58(1):293–301. Available from: <https://doi.org/10.1046/j.1523-1755.2000.00165.x>.
66. Mendy VL, Azevedo MJ, Sarpong DF, et al. The association between individual and combined components of metabolic syndrome and chronic kidney disease among African Americans: the Jackson Heart Study. *PLoS One* 2014;9(7):e101610. Available from: <https://doi.org/10.1371/journal.pone.0101610>.
67. Lea J, Cheek D, Thornley-Brown D, et al. Metabolic syndrome, proteinuria, and the risk of progressive CKD in hypertensive African Americans. *Am J Kidney Dis* 2008;51(5):732–40. Available from: <https://doi.org/10.1053/j.ajkd.2008.01.013>.
68. Lucove J, Vupputuri S, Heiss G, North K, Russell M. Metabolic syndrome and the development of CKD in American Indians: the Strong Heart Study. *Am J Kidney Dis* 2008;51(1):21–8 doi: S0272-6386(07)01309-1 [pii].
69. Thomas G, Sehgal AR, Kashyap SR, Srinivas TR, Kirwan JP, Navaneethan SD. Metabolic syndrome and kidney disease: a systematic review and meta-analysis. *Clin J Am Soc Nephrol* 2011;6(10):2364–73.
70. Yu Z, Rebholz CM, Wong E, et al. Association between hypertension and kidney function decline: the Atherosclerosis Risk in Communities (ARIC) study. *Am J Kidney Dis* 2019;74(3):310–19 doi: S0272-6386(19)30617-1 [pii].
71. Rossing P, Hommel E, Smidt UM, Parving HH. Impact of arterial blood pressure and albuminuria on the progression of diabetic nephropathy in IDDM patients. *Diabetes* 1993;42(5):715–19.
72. de Boer IH, Bangalore S, Benetos A, et al. Diabetes and hypertension: a position statement by the American Diabetes Association. *Diabetes Care* 2017;40(9):1273–84. Available from: <https://doi.org/10.2337/dci17-0026>.
73. Kramer HJ, Townsend RR, Griffin K, et al. KDOQI United States commentary on the 2017 ACC/AHA hypertension guideline. *Am J Kidney Dis* 2019;73(4):437–58 doi: S0272-6386(19)30039-3 [pii].
74. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol* 2018;71(19):e127–248 doi: S0735-1097(17)41519-1 [pii].
75. The SPRINT Research Group. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med* 2015;9. Available from: <https://doi.org/10.1056/NEJMoa1511939> 2015; Published online Nov.
76. ACCORD Study Group, Gerstein HC, Miller ME, et al. Long-term effects of intensive glucose lowering on cardiovascular outcomes. *N Engl J Med* 2011;364(9):818–28. Available from: <https://doi.org/10.1056/NEJMoa1006524>.
77. Coca SG, Ismail-Beigi F, Haq N, Krumholz HM, Parikh CR. Role of intensive glucose control in development of renal end points in type 2 diabetes mellitus: systematic review and meta-analysis intensive glucose control in type 2 diabetes. *Arch Intern Med* 2012;172(10):761–9. Available from: <https://doi.org/10.1001/archinternmed.2011.2230>.

78. MacIsaac RJ, Jerums G, Ekinci EI. Effects of glycaemic management on diabetic kidney disease. *World J Diabetes* 2017;8(5):172–86. Available from: <https://doi.org/10.4239/wjd.v8.i5.172>.
79. Shurraw S, Hemmelgarn B, Lin M, et al. Association between glycemic control and adverse outcomes in people with diabetes mellitus and chronic kidney disease: a population-based cohort study. *Arch Intern Med* 2011;171(21):1920–7. Available from: <https://doi.org/10.1001/archinternmed.2011.537>.
80. ORIGIN trial investigators, Gilbert RE, Mann JF, et al. Basal insulin glargine and microvascular outcomes in dysglycaemic individuals: results of the Outcome Reduction with an Initial Glargine Intervention (ORIGIN) trial. *Diabetologia* 2014;57(7):1325–31. Available from: <https://doi.org/10.1007/s00125-014-3238-4>.
81. Fioretto P, Barzon I, Mauer M. Is diabetic nephropathy reversible? *Diabetes Res Clin Pract* 2014;104(3):323–8. Available from: <https://doi.org/10.1016/j.diabres.2014.01.017>.
82. Fioretto P, Steffes MW, Sutherland DE, Goetz FC, Mauer M. Reversal of lesions of diabetic nephropathy after pancreas transplantation. *N Engl J Med* 1998;339(2):69–75. Available from: <https://doi.org/10.1056/NEJM199807093390202>.
83. Hoy WE, Wang Z, VanBuynder P, Baker PR, Mathews JD. The natural history of renal disease in Australian Aborigines. Part 1. Changes in albuminuria and glomerular filtration rate over time. *Kidney Int* 2001;60(1):243–8.
84. Hoy WE, Wang Z, VanBuynder P, Baker PR, McDonald SM, Mathews JD. The natural history of renal disease in Australian Aborigines. Part 2. Albuminuria predicts natural death and renal failure. *Kidney Int* 2001;60(1):249–56.
85. Ruggenenti P, Perna A, Mosconi L, Pisoni R, Remuzzi G. Urinary protein excretion rate is the best independent predictor of ESRF in non-diabetic proteinuric chronic nephropathies. "Gruppo Italiano di Studi Epidemiologici in Nefrologia" (GISEN). *Kidney Int* 1998;53(5):1209–16.
86. ACCORD Study G, Cushman WC, Evans GW, et al. Effects of intensive blood-pressure control in type 2 diabetes mellitus. *N Engl J Med* 2010;362(17):1575–85.
87. Schenk S, Harber MP, Shrivastava CR, Burant CF, Horowitz JF. Improved insulin sensitivity after weight loss and exercise training is mediated by a reduction in plasma fatty acid mobilization, not enhanced oxidative capacity. *J Physiol (Lond)* 2009;587(20):4949–61 Pt.
88. Park S, Kim YJ, Choi CY, Cho NJ, Gil HW, Lee EY. Bariatric surgery can reduce albuminuria in patients with severe obesity and normal kidney function by reducing systemic inflammation. *Obes Surg* 2018;28(3):831–7. Available from: <https://doi.org/10.1007/s11695-017-2940-y>.
89. Afshinnia F, Wilt TJ, Duval S, Esmaeili A, Ibrahim HN. Weight loss and proteinuria: systematic review of clinical trials and comparative cohorts. *Nephrol Dial Transplant* 2010;25(4):1173–83.
90. Navaneethan SD, Yehner H, Moustarah F, Schreiber MJ, Schauer PR, Beddhu S. Weight loss interventions in chronic kidney disease: a systematic review and meta-analysis. *Clin J Am Soc Nephrol* 2009;4(10):1565–74. Available from: <https://doi.org/10.2215/CJN.02250409>.
91. Rutkowski JM, Wang ZV, Park AS, et al. Adiponectin promotes functional recovery after podocyte ablation. *J Am Soc Nephrol* 2013;24(2):268–82.
92. Kim Y, Park CW. Mechanisms of adiponectin action: implication of adiponectin receptor agonism in diabetic kidney disease. *Int J Mol Sci* 2019;20(7) 10.3390/ijms20071782. doi: E1782 [pii].
93. Qiao L, Shao J. SIRT1 regulates adiponectin gene expression through Foxo1-C/enhancer-binding protein alpha transcriptional complex. *J Biol Chem* 2006;281(52):39915–24.
94. Mattagajasingh I, Kim CS, Naqvi A, et al. SIRT1 promotes endothelium-dependent vascular relaxation by activating endothelial nitric oxide synthase. *Proc Natl Acad Sci U S A* 2007;104(37):14855–60.
95. Cheng KK, Lam KS, Wang Y, et al. Adiponectin-induced endothelial nitric oxide synthase activation and nitric oxide production are mediated by APPL1 in endothelial cells. *Diabetes* 2007;56(5):1387–94 doi: db06-1580 [pii].
96. Knowler WC, Barrett-Connor E, Fowler SE, et al. Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. *N Engl J Med* 2002;346(6):393–403. Available from: <https://doi.org/10.1056/NEJMoa012512>.
97. Diabetes Prevention Program Research Group. Long-term effects of lifestyle intervention or metformin on diabetes development and microvascular complications over 15-year follow-up: the Diabetes Prevention Program Outcomes Study. *Lancet Diabetes Endocrinol* 2015;3(11):866–75. Available from: [https://doi.org/10.1016/S2213-8587\(15\)00291-0](https://doi.org/10.1016/S2213-8587(15)00291-0).
98. Lindstrom J, Ilanne-Parikka P, Peltonen M, et al. Sustained reduction in the incidence of type 2 diabetes by lifestyle intervention: follow-up of the Finnish Diabetes Prevention Study. *Lancet* 2006;368(9548):1673–9 doi: S0140-6736(06)69701-8 [pii].
99. Li G, Zhang P, Wang J, et al. The long-term effect of lifestyle interventions to prevent diabetes in the China Da Qing Diabetes Prevention Study: a 20-year follow-up study. *Lancet* 2008;371(9626):1783–9. Available from: [https://doi.org/10.1016/S0140-6736\(08\)60766-7](https://doi.org/10.1016/S0140-6736(08)60766-7).
100. Salas-Salvado J, Bullo M, Babio N, et al. Reduction in the incidence of type 2 diabetes with the Mediterranean diet: results of the PREDIMED-Reus nutrition intervention randomized trial. *Diabetes Care* 2011;34(1):14–19. Available from: <https://doi.org/10.2337/dc10-1288>.
101. Bloomfield HE, Koeller E, Greer N, MacDonald R, Kane R, Wilt TJ. Effects on health outcomes of a Mediterranean diet with no restriction on fat intake: a systematic review and meta-analysis. *Ann Intern Med* 2016;165(7):491–500 doi: 10.7326/M16-0361.
102. American Diabetes Association. 5. Prevention or delay of type 2 diabetes. *Diabetes Care* 2017;40(1):S44–7 Suppl.
103. Buse JB, DeFronzo RA, Rosenstock J, et al. The primary glucose-lowering effect of metformin resides in the gut, not the circulation: results from short-term pharmacokinetic and 12-week dose-ranging studies. *Diabetes Care* 2016;39(2):198–205. Available from: <https://doi.org/10.2337/dc15-0488>.
104. Sarafidis PA, Stafylas PC, Georgianos PI, Saratzis AN, Lasaridis AN. Effect of thiazolidinediones on albuminuria and proteinuria in diabetes: a meta-analysis. *Am J Kidney Dis* 2010;55(5):835–47. Available from: <https://doi.org/10.1053/jajkd.2009.11.013>.
105. Chatterjee S, Ghosal S, Chatterjee S. Glucagon-like peptide-1 receptor agonists favorably address all components of metabolic syndrome. *World J Diabetes* 2016;7(18):441–8. Available from: <https://doi.org/10.4239/wjd.v7.i18.441>.
106. Marso SP, Bain SC, Consoli A, et al. Semaglutide and cardiovascular outcomes in patients with type 2 diabetes. *N Engl J Med* 2016;375(19):1834–44. Available from: <https://doi.org/10.1056/NEJMoa1607141>.
107. Rameshrad M, Razavi BM, Ferns GAA, Hosseinzadeh H. Pharmacology of dipeptidyl peptidase-4 inhibitors and its use in the management of metabolic syndrome: a comprehensive review on drug repositioning. *Daru* 2019;27(1):341–60. Available from: <https://doi.org/10.1007/s40199-019-00238-7>.
108. Cooper ME, Perkovic V, McGill JB, et al. Kidney disease end points in a pooled analysis of individual patient-level data from a large clinical trials program of the dipeptidyl peptidase 4 inhibitor linagliptin in type 2 diabetes. *Am J Kidney Dis* 2015;66(3):441–9. Available from: <https://doi.org/10.1053/ajkd.2015.03.024>.

109. Vergara A, Jacobs-Cacha C, Soler MJ. Sodium-glucose cotransporter inhibitors: beyond glycaemic control. *Clin Kidney J* 2019;12(3):322–5. Available from: <https://doi.org/10.1093/ckj/sfz019>.
110. Neuen BL, Young T, Heerspink HJL, et al. SGLT2 inhibitors for the prevention of kidney failure in patients with type 2 diabetes: a systematic review and meta-analysis. *Lancet Diabetes Endocrinol* 2019; doi: S2213-8587(19)30256-6 [pii].
111. Perkovic V, Jardine MJ, Neal B, et al. Canagliflozin and renal outcomes in type 2 diabetes and nephropathy. *N Engl J Med* 2019;380(24):2295–306. Available from: <https://doi.org/10.1056/NEJMoa1811744>.
112. Turner-McGrievy G, Mandes T, Crimarco A. A plant-based diet for overweight and obesity prevention and treatment. *J Geriatr Cardiol* 2017;14(5):369–74. Available from: <https://doi.org/10.11909/j.issn.1671-5411.2017.05.002>.
113. Koh-Banerjee P, Franz M, Sampson L, et al. Changes in whole-grain, bran, and cereal fiber consumption in relation to 8-y weight gain among men. *Am J Clin Nutr* 2004;80(5):1237–45 doi: 80/5/1237 [pii].
114. Tucker LA, Thomas KS. Increasing total fiber intake reduces risk of weight and fat gains in women. *J Nutr* 2009;139(3):576–81. Available from: <https://doi.org/10.3945/jn.108.096685>.
115. Barnard ND, Levin SM, Yokoyama Y. A systematic review and meta-analysis of changes in body weight in clinical trials of vegetarian diets. *J Acad Nutr Diet* 2015;115(6):954–69. Available from: <https://doi.org/10.1016/j.jand.2014.11.016>.
116. Lattimer JM, Haub MD. Effects of dietary fiber and its components on metabolic health. *Nutrients* 2010;2(12):1266–89. Available from: <https://doi.org/10.3390/nu2121266>.
117. Orlich MJ, Singh PN, Sabate J, et al. Vegetarian dietary patterns and mortality in Adventist Health Study 2. *JAMA Intern Med* 2013;173(13):1230–8. Available from: <https://doi.org/10.1001/jamainternmed.2013.6473>.
118. Lin J, Fung TT, Hu FB, Curhan GC. Association of dietary patterns with albuminuria and kidney function decline in older white women: a subgroup analysis from the Nurses' Health Study. *Am J Kidney Dis* 2011;57(2):245–54.
119. Yuzbashian E, Asghari G, Mirmiran P, Hosseini FS, Azizi F. Associations of dietary macronutrients with glomerular filtration rate and kidney dysfunction: Tehran lipid and glucose study. *J Nephrol* 2015;28(2):173–80. Available from: <https://doi.org/10.1007/s40620-014-0095-7>.
120. Goraya N, Simoni J, Jo CH, Wesson DE. A comparison of treating metabolic acidosis in CKD stage 4 hypertensive kidney disease with fruits and vegetables or sodium bicarbonate. *Clin J Am Soc Nephrol: CJASN* 2013;8(3):371–81.
121. Gluba-Brzozka A, Franczyk B, Rysz J. Vegetarian diet in chronic kidney disease—a friend or foe. *Nutrients* 2017;9(4) 10.3390/nu9040374. doi: E374 [pii].